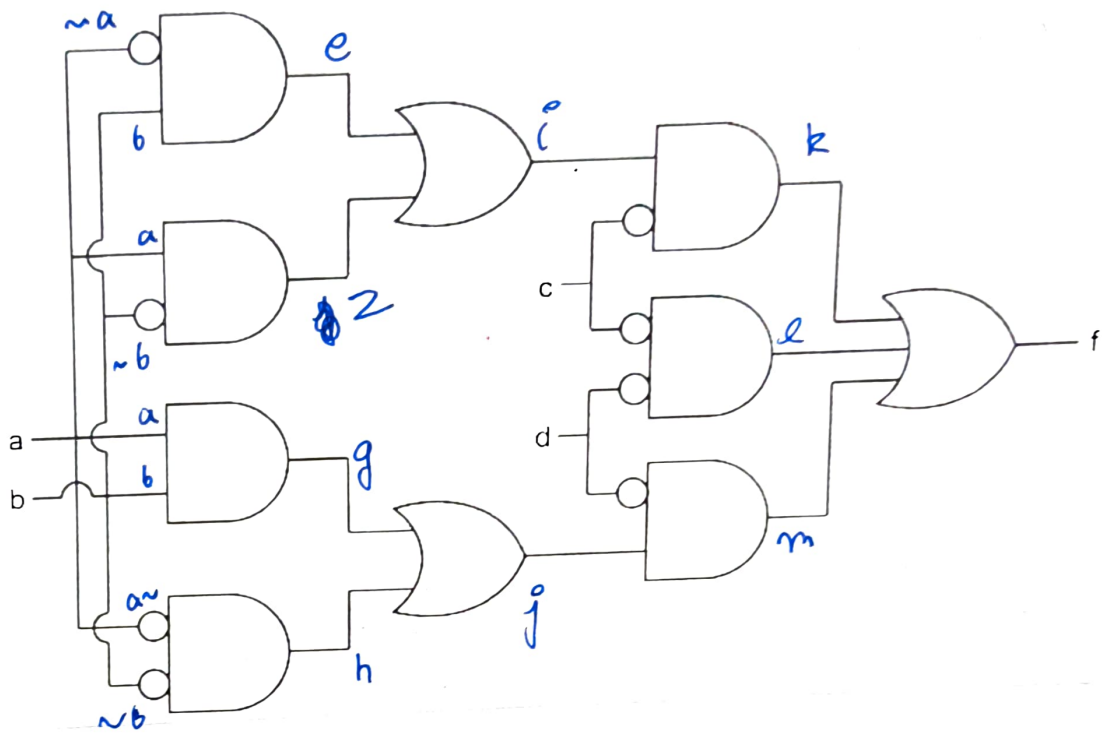


- Write a Verilog code for the digital circuit shown below using data flow modelling. (8 Marks)



- Explain the difference between simulations after logic synthesis stage and after routing stage of FPGA design flow. (2 marks)

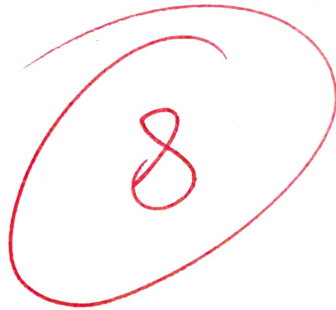
Q1. module circuit (input a,b, ~~output~~ c,d, output f);
 wire e, ~~g~~^z, h, i, j, k, l, m;
 assign e = ($\sim a$) & (b); ✓
 assign ~~z~~^z = (a) & ($\sim b$); ✓
 assign g = (a) & (b); ✓
 assign h = ($\sim a$) & ($\sim b$); ✓
 assign i = e | ~~z~~^z; ✓
 assign j = g | h; ✓
 assign k = ($\sim c$) & (i); ✓
 assign l = ($\sim c$) & ($\sim d$); ✓

assign m = (vd) & (j);

assign j = k | l | m;

~~endmodule~~

endmodule



Q2.

1) Post logic synthesis ^{run}: Simulation done after tool creates gate level representation of verilog code. This checks if code runs on gate level rep in the tool itself not on the board (FPGA)

2

2) Post routing ^{stage} ~~synthesis~~ ^{run}: Simulation done after routing look up tables on the FPGA, these routers add ^{based on distance} additional delays hence this sim ~~run~~ checks if functionality remains same after additional delays.